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CS101

When a single dictionary isn't enough

Today
Welcome to your next lesson: When a single dictionary isn't enough.

I'm getting your personalized lesson ready — based on what you've learned, what's ahead, and how you learn best.

Step by step, you’re moving forward.

Imagine you’re tracking your own video game characters in a notebook 🎮

You might make a little table like this:

name	level	class
Aria	5	mage
Bron	3	warrior
Kira	7	rogue

Each row is one character. Each column is a field: name, level, class.

In Python, you already know how to store one row as a dictionary, like:

Python

```
aria = {
    "name": "Aria",
    "level": 5,
    "class": "mage"
}
```

This is “one record” / “one row”.

Quick check to place you:

Have you already created dictionaries like aria before (one thing with several fields), or does that still feel new?

I've done it before

Nice 👍 that helps us go a bit faster.

You already know:

Python

```
aria = {
```

```
    "name": "Aria",
    "level": 5,
    "class": "mage"
}
```

This is one character = one dictionary.

Now imagine you want to keep many characters together: aria, bron, kira.

We need something that can hold multiple items in order → that's a list in Python: [...].

So the pattern we're aiming for is:

one character → { ... }

many characters → [{ ... }, { ... }, { ... }]

Does it make sense in words that we'll put several dictionaries inside a list to hold many characters?

If yes, I'll show you a concrete Python example next.

Yes, that makes sense

Great 👌

Let's now see a full list of dictionaries for those three characters.

Visually, think:

Text

```
[      a list (many things)
  { ... },   first character
  { ... },   second character
  { ... }    third character
]
```

In Python, that looks like:

Python

```
characters = [
    {"name": "Aria", "level": 5, "class": "mage"},
    {"name": "Bron", "level": 3, "class": "warrior"},
    {"name": "Kira", "level": 7, "class": "rogue"}
]

print(characters)
```

Key structure:

Outer [ ] → the list of all characters
Inner { } → each single character dictionary
Does this structure match how you think about the table of rows?

If anything feels off (especially the brackets), tell me which part looks confusing.

Yes, it matches

Perfect 🎯 Since the idea makes sense in words and in code, let's have you build your own.

Step by step, in the code editor:

Create a variable called pets.
Set it equal to a list [...].
Inside that list, create 3 dictionaries, each with the same keys:
"name"
"species"
"age"
Then print(pets).

Shape to aim for (don’t copy this, just the shape):

Python

```
pets = [
    {...},
    {...},
    {...}
]

print(pets)
```

Go ahead and write this in the editor, then hit Submit code (not just Run).

I’ll check your structure and help fix anything that’s off 👀

Python

```
pets = [
    {"name": "Hulk", "species": "Abnormal Human", "age": 45},
    {"name": "gnobby", "species": "gnome", "age": 220},
    {"name": "Elveryn", "species": "elven", "age": 650}

]

print(pets)
```

Nice work, Ron 🎉 This is exactly the pattern we want.

You’ve got:

Outer list: [...] holding all pets
Inner dictionaries: {...} each with the same keys: "name", "species", "age"

So in “table” form, your data is:

```
name    species age
Hulk    Abnormal Human  45
gnobby  gnome   220
Elveryn elven   650
```

One row → one dictionary.

All rows together → one list of dictionaries.

Quick check, in your own words:

Why is a list of dictionaries a good fit when you have “rows with the same columns” like this?

(Short answer is fine.)

it fits a database style like row types with like columns the sam every time

Exactly 💯 — you nailed the idea.

You’ve got:

Each dictionary = one “row type” with the same columns/fields (name, species, age)
The list = the whole “table” of those rows, in order

That’s why this pattern is so natural for database-ish or spreadsheet-style data.

Want to do one more tiny practice to lock it in, or do you feel solid and ready to move on from lists of dictionaries?

we can move ahead think i got it

Love it. You really do have it 👍

Quick recap before we switch lessons:

You can build a list of dictionaries like:

```
pets = [
    {"name": "Hulk", "species": "Abnormal Human", "age": 45},
    {"name": "gnobby", "species": "gnome", "age": 220},
    {"name": "Elveryn", "species": "elven", "age": 650}
]
```

You understand the mental model:
One row → one dictionary (same keys each time)
Many rows → one list of those dictionaries
You connected it to database / table style data, which is exactly how professionals think about this pattern.
You’re ready for what’s next 🚀
When you’re set, click Next Lesson in the Learn tab, and I’ll keep building on this with you.

```
pets = [
    {"name": "Hulk", "species": "Abnormal Human", "age": 45},
    {"name": "gnobby", "species": "gnome", "age": 220},
    {"name": "Elveryn", "species": "elven", "age": 650}

]

print(pets)
```